

Education

Sant'Anna School of Advanced Studies

Sep. 2024 – Current

Advanced MSc (EQF 8): Robotics, Pisa, Italy

- Awarded 2-Year Merit Full Scholarship (top 2% of applicants).
- Current GPA: 30/30.

University of Pisa

Sep. 2024 – Jul. 2026

MSc in Artificial Intelligence and Data Engineering, Pisa, Italy

- Final Grade: 110 Cum Laude (GPA: 29.93/30).
- Thesis: "Development and Evaluation of a Domain-Adversarial Transformer for Subject-Invariant Physiological Signals Analysis". Supervisor: M. Cimino.

Budapest University of Technology and Economics

Dec. 2024

Seasonal School: Challenge-based human-inspired computer vision, Budapest, Hungary

University of Padua

Oct. 2021 – Jul. 2024

BSc in Computer Engineering, Padua, Italy

- DECAMP Course in Applied Web Application Security (Metropolia UAS, Finland).
- Final Grade: 110 Cum Laude (GPA: 29.84/30).
- Thesis: "A deep learning approach for the analysis of calcium signal in characterizing the biological response of plants to external stimuli". Supervisor: E. Menegatti.

Scientific High School G.B. Ferrari

Sep. 2016 – Jul. 2021

High School Diploma in Applied Sciences, Este, Italy

- Final Grade: 100 Cum Laude (GPA: 9.2/10).

Research Experience

SLAC National Accelerator Laboratory (Stanford University)

Jul. 2026 – Sep. 2026

Visiting Scientist - Vernieri Group, Menlo Park, CA

- Developing an agentic AI system to automate the optimization and design of Future Circular Collider (FCC) particle detectors. Supervisor: C. Vernieri.

Mila (Quebec Artificial Intelligence Institute)

Mar. 2026 – Jun. 2026

Visiting Researcher - sciNeurotech Lab (PolyMTL), Montréal, Canada

- Developed CNN surrogates to replace computationally expensive online Gaussian Process updates. Supervisor: M. Bonizzato.

Research Center "E. Piaggio"

Oct. 2025 – Jul. 2026

Student Researcher - MLPI Lab, Pisa, Italy

- Framing R-peak prediction from EEG as a segmentation problem via distance-to-peak estimation, evaluating multi-scale CNNs and autoregressive models with a focus on domain generalization. Supervisor: M. Cimino.

Monasterio Foundation

Sep. 2025 – Jan. 2026

Student Researcher - PERCRO Lab, Pisa, Italy

- Cardiac amyloidosis classification from ECGs using Transformers, with interpretability via Grad-CAM and Attention Rollout. Supervisor: C.A. Avizzano.

Sant'Anna School of Advanced Studies

Feb. 2025 – Oct. 2025

Student Researcher - NeuroRobotic Touch Lab and TeCIP Institute, Pisa, Italy

- Explainable classification of tumor-mimicking materials from force/position data. Supervisor: C.M. Oddo.
- Efficient federated learning via dynamic training-precision adjustment. Supervisor: E. Paolini.

RFX Consortium and CNR

Nov. 2023 – Feb. 2024

Internship - Nuclear Fusion and Plasma Research, Padua, Italy

- Investigated autoencoders and VAEs for temporal analysis of plant stress responses after irrigation with plasma-activated water (PAW). Supervisor: V. Antoni.

Work Experience

UP caeli via

Mar. 2024 – Jul. 2024

Computer Engineer – Part Time (24h/week), Padua, Italy

- Software integration for drones (QGC and ArduPilot) using C++ and Qt/QML (GeoFencing, RTK and RTSP).

Formula Student Team Project

Oct. 2022 – Aug. 2024

Embedded Software Developer (+10h/week), Padua, Italy

- Member of a 50-person university racing team, developing embedded systems (ATmega32/64, TI F2837, Arduino) and C control algorithms for power management, torque vectoring, and regenerative braking.

Publications & Presentations

- **AI-Driven Optimization of Detector Design for the FCC-ee.** 4th US FCC Meeting, SLAC, Sep. 8-11, 2026. Presented research on multifidelity Bayesian optimization and differentiable simulation surrogates for efficient design optimization for the proposed FCC-ee at CERN.
- **ResCTrans: an Explainable Amyloidosis Classifier using Temporal Transformers from ECG Signals.** Submitted to *IEEE Journal of Biomedical and Health Informatics*, Jul. 9, 2026.

Selected Honours and Awards

- | | |
|--|-----------|
| • Zegna Scholarship: Scholarship awarded for academic excellence and merit | Jun. 2026 |
| • Mitacs Globalink Research Scholarship: Scholarship supporting a visiting stay at Mila | Mar. 2026 |
| • Mille e una Lode Merit Awards: Scholarships awarded to the top 3% of students (UniPD) | 2022–2024 |
| • INPS Scholarships: National scholarships in recognition of academic excellence | 2018–2024 |
| • National Bronze Medal, Physics Olympiad: Association Teaching of Physics | May 2021 |
| • National First Place, Chemistry Olympiad: Italian Chemistry Society | Jun. 2020 |
| • National Bronze Medal, EUSO: European Olympiad of Experimental Science | Mar. 2019 |

Memberships

- | | |
|---|---------------------|
| • LeadTheFuture Mentee: Acceptance < 15% for leading STEM mentorship | Oct. 2025 – Current |
| • ISSNAF Mentee: Selective mentorship for research opportunities in the US | Dec. 2025 – Current |

Languages

- **Italian:** Native
- **English:** C1 (TOEFL iBT 104/120)